

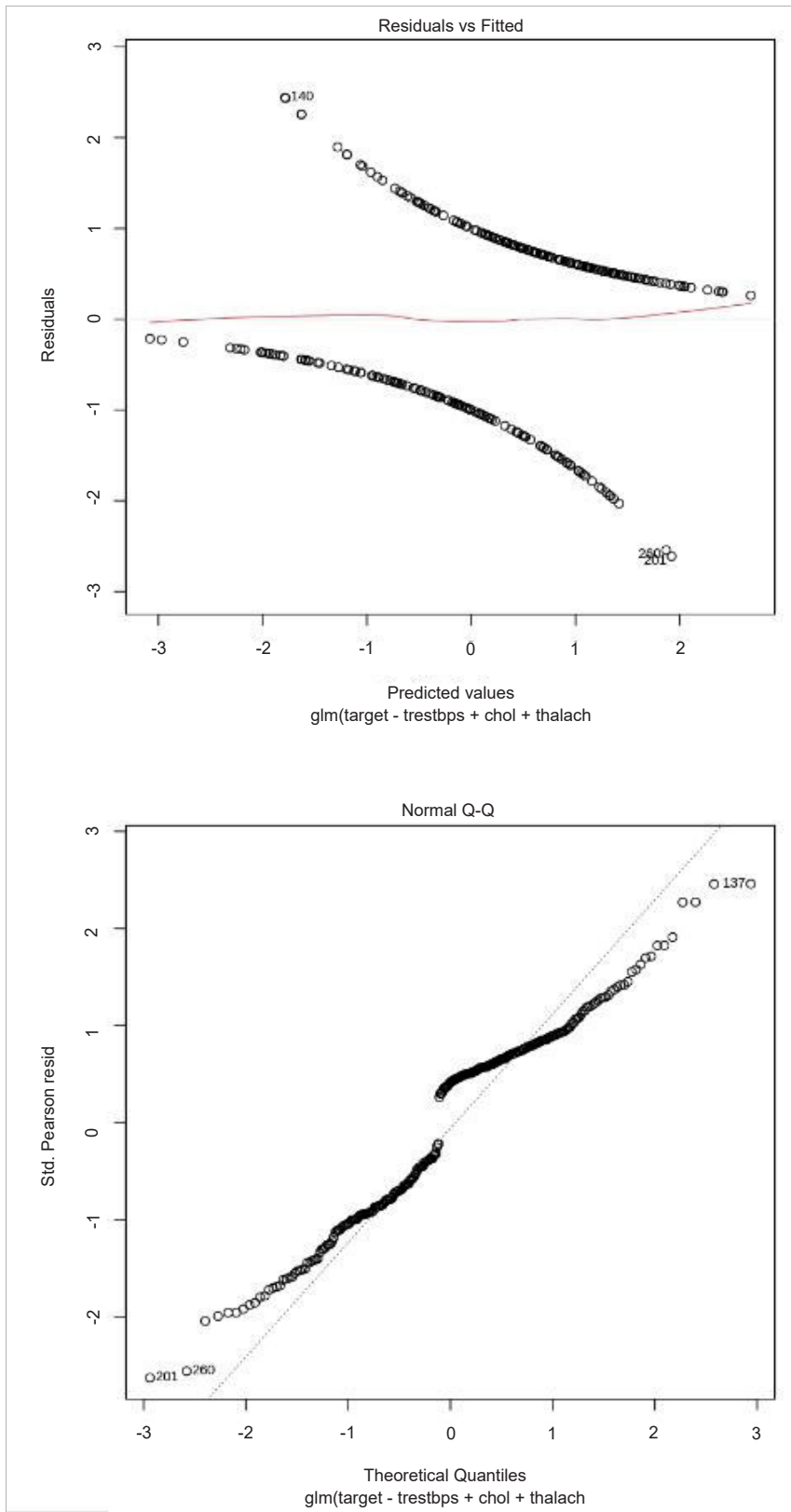


Fig. 7: Graphical representation of residuals vs filtered

To construct the GLM, type:

```
print("Constructing the Logistic
regression Model")
glm(target~ trestbps+ restecg + fbs,
data = data, family=binomial())
model <- glm(target~trestbps+ chol +
thalach, data = data, family=binomial())
plot(model)
```

The outcomes of the construction of the logistic regression model are illustrated in Fig. 6.

Use the following code to construct the decision tree model:

```
print("Construction of the Decision Tree
Model") mtree <- rpart(target ~ trestbps
+ chol + thalach, data = train,
method="class", control = rpart.
control(minsplit = 20, minbucket = 7,
maxdepth = 10, usesurrogate = 2, xval
=10))
```

Now plot the decision tree for the dataset, using 'mtree':

```
print("Plotting the Decision Tree")
plot(mtree)
text(mtree)
par(xpd = NA, mar = rep(0.7, 4))
plot(mtree, compress = TRUE)
text(mtree, cex = 0.7, use.n = TRUE,
fancy = FALSE, all = TRUE) prp(mtree,
faclen = 0, box.palette = "Reds", cex =
0.8, extra = 1)
```

Next, construct the random forest model:

```
rf <- randomForest(target ~ trestbps +
oldpeak + cp, data = data)
```

View the forest results:

```
print("Random Forest Results:")
print(rf)
```

The importance of each predictor can be gauged by:

```
print("Importance of each predictor:")
print(importance(rf,type = 2))
```

To plot the random forest, type:

```
plot(rf)
```

The graphical outcomes of the regression model are depicted in Figs 7 and 8.

This IoT-based healthcare application for predicting the possibility of a heart attack using GLMs, random forests, and decision trees in R holds significant promise